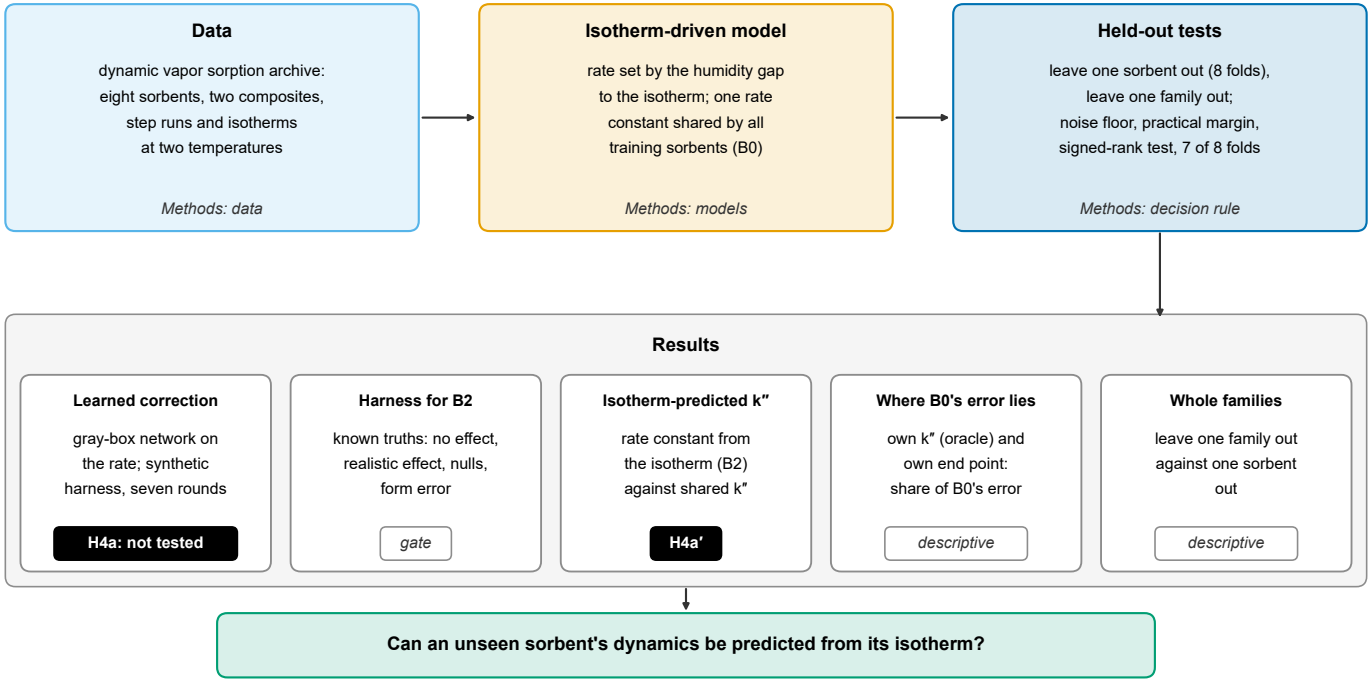

Data
dynamic vapor sorption archive:
eight sorbents, two composites,
step runs and isotherms
at two temperatures
Methods: data
Isotherm-driven model
rate set by the humidity gap
to the isotherm; one rate
constant shared by all
training sorbents (B0)
Methods: models
Held-out tests
leave one sorbent out (8 folds),
leave one family out;
noise floor, practical margin,
signed-rank test, 7 of 8 folds
Methods: decision rule
Results
Learned correction
gray-box network on
the rate; synthetic
harness, seven rounds
H4a: not tested
Harness for B2
known truths: no effect,
realistic effect, nulls,
form error
gate
Isotherm-predicted k″
rate constant from
the isotherm (B2)
against shared k″
H4a′
Where B0's error lies
own k″ (oracle) and
own end point:
share of B0's error
descriptive
Whole families
leave one family out
against one sorbent
out
descriptive
Can an unseen sorbent's dynamics be predicted from its isotherm?